



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRK PETROLEUM AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

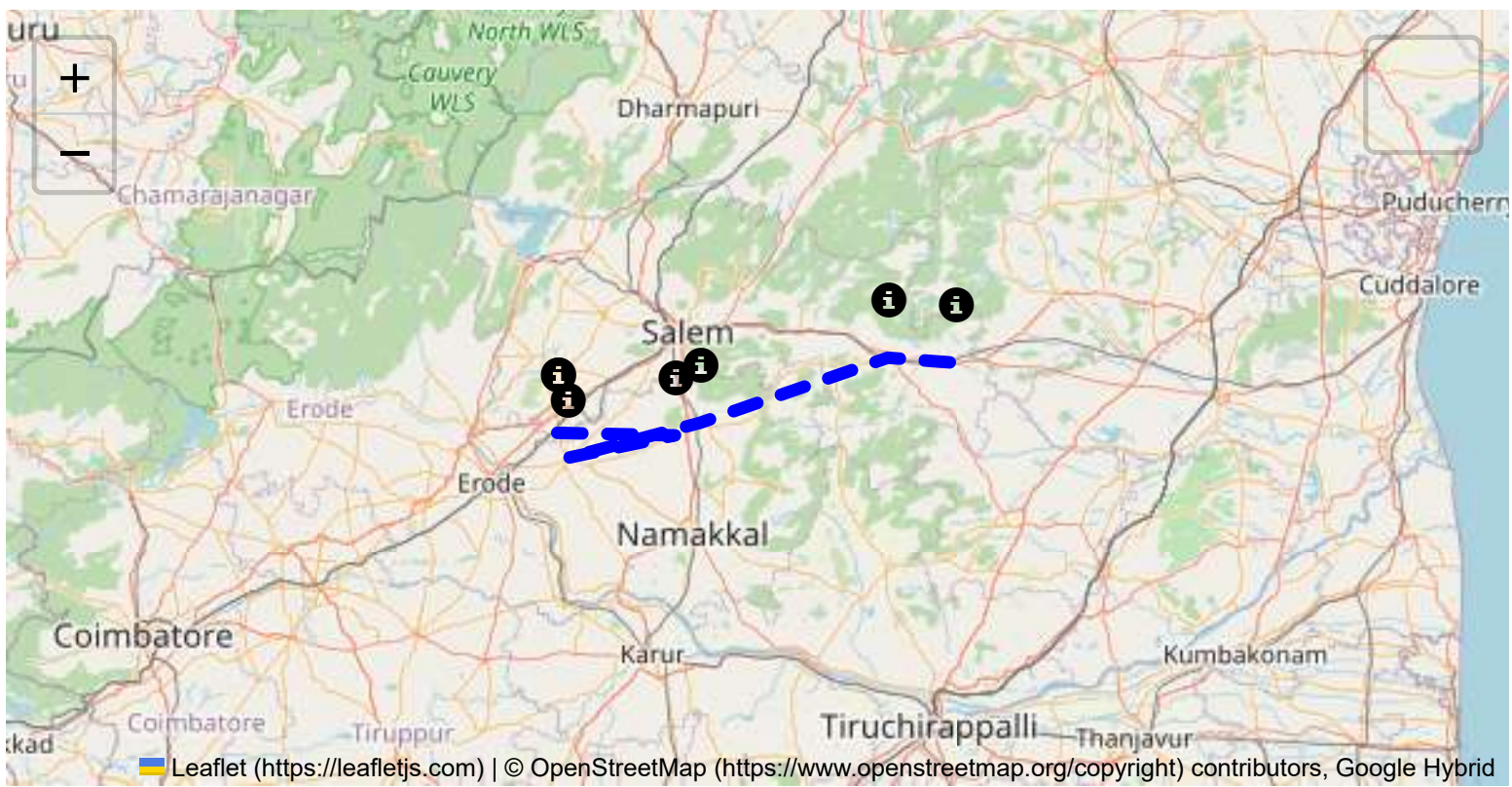
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

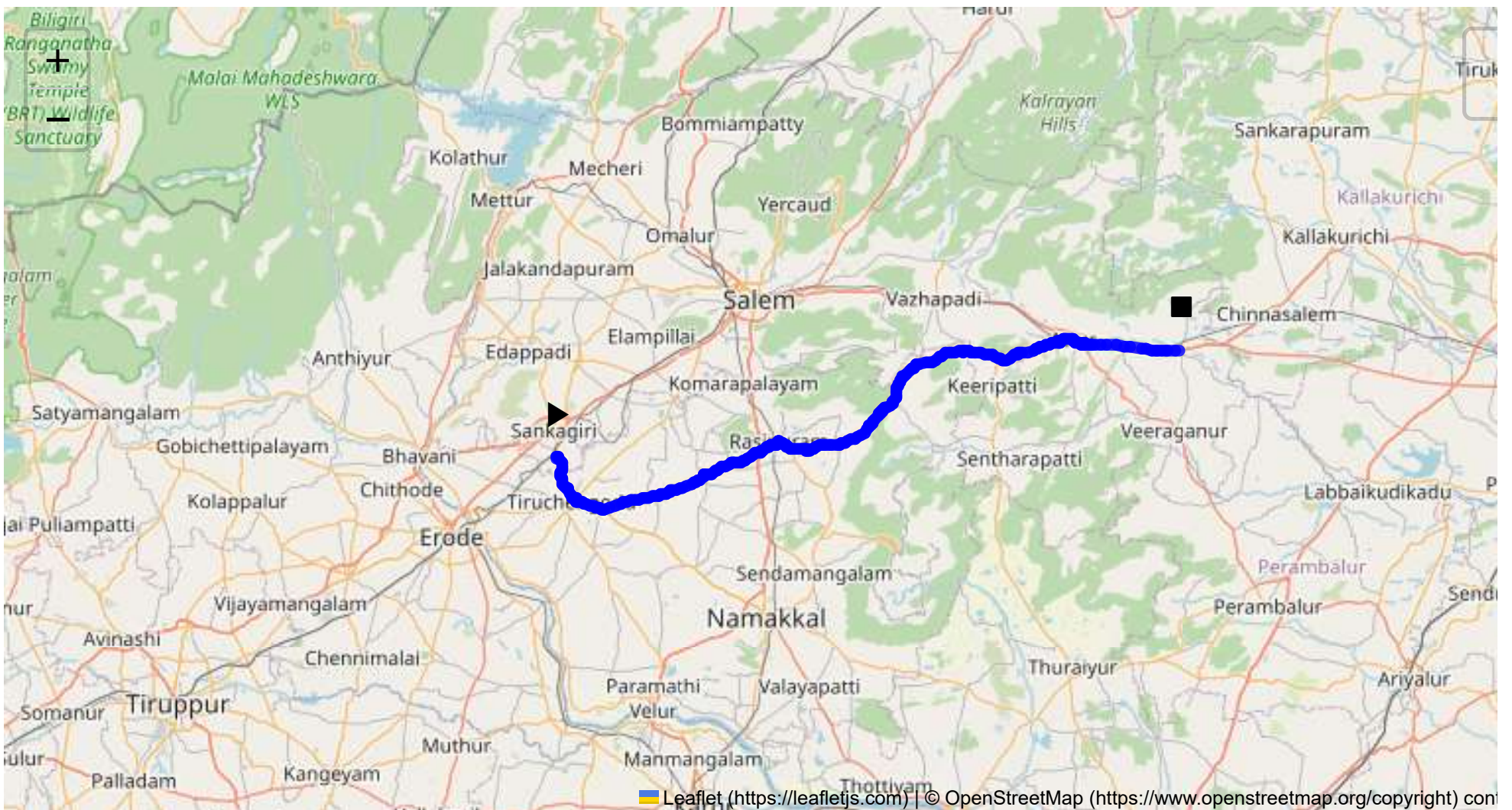
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 113.92 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (11.585986, 78.744957)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Thalaivasal via specified points spans approximately 113.92 kilometers. It navigates through a mix of rural and semi-urban areas, primarily following state highways and local roads. Key waypoints include Puthur, Pillanallur, Tiruchengode, Rasipuram, and Shakti Complex. The journey features a combination of straight driving on highways and interactions with local traffic near towns.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. Conditions vary from dry and hot in the summer months to heavy rains during the monsoon season (June to September). Monsoon can lead to potential flooding and slippery roads, particularly in the rural and less developed road segments. Summer heat may also affect vehicle performance and driver fatigue.

3. Analysis of Traffic Patterns

Traffic intensity is typically higher during peak hours, from 8-10 AM and 5-7 PM, especially around towns like Tiruchengode and Rasipuram. These areas can experience congestion due to local commuting and market days. The highway segments are relatively smoother but can have intermittent heavy vehicle traffic.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route. State highways are generally well-maintained, while smaller roads, especially in rural areas, might have potholes or uneven surfaces. Adequate signage is generally present on highways but may be limited on smaller roads.

5. Suggestions for Alternative Routes for Emergencies

If portions of the route are blocked or pose risks, one potential alternative is to use National Highway 44, which is broader and generally better maintained, although it may demand a longer travel distance. Moreover, exploring local state highways that run parallel to the primary route can provide some essential detours.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu adheres to national safety standards and regulations for the transport of hazardous materials. This includes restrictions on transport timings (to avoid peak hours in congested areas) and mandates the use of proper safety signage on vehicles.

7. Overview of Historical Incidents

Though specific incident data is not readily available, Tamil Nadu's records of accidents often include heavy vehicles. Key concerns involve truck rollovers on curves, particularly during adverse weather, and collisions at busy intersections. Monitoring local news archives or transport authority updates for specific historical data would be advisable.

8. Environmental Considerations and Sensitive Areas

En route, environmentally sensitive areas include agricultural zones and water bodies near Rasipuram. Ensure not to spill or leave residues in these zones as it can harm local ecosystems.

9. Analysis of Communication Coverage

The route has reasonably good mobile network coverage in urban and semi-urban areas. However, there may be pockets of weak signal in rural stretches between major towns. It is advisable to have communication equipment that can work in low-signal conditions.

10. Estimated Emergency Response Times

Urban segments like Rasipuram have quicker emergency response times, approximately 20-30 minutes. In contrast, rural areas could take up to an hour or more, depending on the accessibility and distance from emergency facilities.

11. Overall Summary of Risk Assessment

Overall, the journey from Sangagiri to Thalaivasal presents moderate risks. Major concerns are weather-related road conditions and traffic congestion in towns. Preparedness for emergencies like vehicle breakdowns and clear communication plans are essential. Following regulations and staying updated on road conditions will further mitigate risks. Further coordination with local emergency services might be necessary to ensure safety and compliance.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44022, 77.87546	15 KM/Hr
3	Turn	Medium	11.43107, 77.88171	30 KM/Hr
4	Turn	Medium	11.39976, 77.88382	30 KM/Hr
5	Turn	Medium	11.39690, 77.88852	30 KM/Hr
6	Turn	Medium	11.38388, 77.89544	30 KM/Hr
7	Turn	Medium	11.38386, 77.89881	30 KM/Hr
8	Turn	High	11.38190, 77.90254	15 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Turn	Medium	11.37820, 77.90198	30 KM/Hr
11	Turn	Medium	11.41354, 78.08075	30 KM/Hr
12	Turn	High	11.41447, 78.08063	15 KM/Hr
13	Turn	High	11.41475, 78.08087	15 KM/Hr
14	Turn	High	11.43257, 78.12096	15 KM/Hr
15	Turn	High	11.45597, 78.18073	15 KM/Hr
16	Turn	Medium	11.45728, 78.18115	30 KM/Hr
17	Turn	Medium	11.46015, 78.18459	30 KM/Hr
18	Blind Spot	Blind Spot	11.46034, 78.18528	10 KM/Hr
19	Turn	High	11.45909, 78.18553	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
20	Turn	Medium	11.44994, 78.22115	30 KM/Hr
21	Turn	High	11.44925, 78.22288	15 KM/Hr
22	Turn	High	11.44813, 78.22305	15 KM/Hr
23	Turn	Medium	11.44767, 78.22372	30 KM/Hr
24	Turn	Medium	11.47202, 78.30442	30 KM/Hr
25	Turn	Medium	11.49812, 78.32398	30 KM/Hr
26	Turn	Medium	11.50497, 78.33079	30 KM/Hr
27	Turn	Medium	11.50769, 78.33291	30 KM/Hr
28	Turn	Medium	11.50860, 78.33951	30 KM/Hr
29	Turn	Medium	11.51254, 78.34450	30 KM/Hr
30	Turn	Medium	11.51907, 78.34826	30 KM/Hr
31	Turn	Medium	11.54803, 78.35598	30 KM/Hr
32	Turn	Medium	11.58161, 78.41769	30 KM/Hr
33	Turn	Medium	11.58329, 78.44358	30 KM/Hr
34	Turn	Medium	11.57844, 78.48399	30 KM/Hr
35	Turn	Medium	11.57468, 78.49452	30 KM/Hr
36	Turn	Medium	11.57147, 78.49830	30 KM/Hr
37	Turn	Medium	11.57097, 78.50062	30 KM/Hr
38	Turn	Medium	11.57848, 78.51320	30 KM/Hr
39	Turn	Medium	11.57845, 78.51539	30 KM/Hr
40	Turn	Medium	11.58692, 78.54587	30 KM/Hr
41	Turn	Medium	11.59082, 78.55593	30 KM/Hr
42	Turn	Medium	11.59228, 78.55728	30 KM/Hr
43	Turn	High	11.59744, 78.57347	15 KM/Hr
44	Turn	High	11.59550, 78.57463	15 KM/Hr
45	Turn	High	11.59533, 78.57596	15 KM/Hr
46	Turn	High	11.59889, 78.57708	15 KM/Hr
47	Turn	Medium	11.59974, 78.58060	30 KM/Hr
48	Turn	High	11.59965, 78.59643	15 KM/Hr
49	Turn	Medium	11.59891, 78.59656	30 KM/Hr
50	Turn	Medium	11.59635, 78.60310	30 KM/Hr
51	Turn	Medium	11.59377, 78.60594	30 KM/Hr
52	Turn	High	11.59371, 78.61015	15 KM/Hr
53	Turn	High	11.59503, 78.61074	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
54	Turn	Medium	11.59314, 78.62100	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
12	hospital	Sudha Hospital	11.4555773, 78.1760125	30 km/h	Medium
13	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
14	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
15	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
16	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
17	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
18	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
19	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium
20	hospital	Manickkam Hospital	11.4568944, 78.1926061	30 km/h	Medium
21	hospital	R K Eye Care Center	11.4589079, 78.1913596	30 km/h	Medium
22	hospital	Snathi Hospital	11.4564245, 78.1932485	30 km/h	Medium
23	hospital	Sugam Hospital,Rasipuram	11.4553007, 78.1936221	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
25	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
26	clinic	Dr. Saravanakumar Clinic	11.5842002, 78.4441223	30 km/h	Medium
27	hospital	Government Hospital	11.572472, 78.497407	30 km/h	Medium
29	hospital	Abi Hospital	11.59319, 78.560129	30 km/h	Medium
32	hospital	Sundaram Multi Speciality Hospital	11.600633, 78.5948143	30 km/h	Medium
33	clinic	RK Clinic	11.59969, 78.5941386	30 km/h	Medium
34	hospital	Kalyani Memorial Hospital	11.599983, 78.596223	30 km/h	Medium
35	clinic	Mother's Ortho Clinic	11.6002438, 78.594053	30 km/h	Medium
36	hospital	Chiranjeevi Hospital	11.5998283, 78.594309	30 km/h	Medium
37	clinic	Sri Vinayaga Physiotherapy Clinic	11.598134, 78.597401	30 km/h	Medium
38	hospital	Sri Andavhar Nursing Home	11.6010351, 78.5974614	30 km/h	Medium
39	clinic	Attur ENT and Dental Care Centre	11.5991073, 78.5962503	30 km/h	Medium
40	clinic	SS Clinic	11.5975084, 78.5979036	30 km/h	Medium
41	hospital	Arun Hospital	11.599848, 78.596343	30 km/h	Medium
42	clinic	Bawaa Clinic	11.597934, 78.597143	30 km/h	Medium
43	clinic	Ramakrishna Nursing Home	11.5988349, 78.5988615	30 km/h	Medium
44	hospital	Gokulam Hospital	11.597356, 78.598403	30 km/h	Medium
45	hospital	Kumar Eye Hospital	11.5978065, 78.5984006	30 km/h	Medium
46	hospital	Anugraha Hospital	11.5963566, 78.5941161	30 km/h	Medium
47	hospital	Geeth Ragunath Hospital	11.5954889, 78.5955383	30 km/h	Medium
48	hospital	N.S. Kumudham Hospital	11.59626, 78.599512	30 km/h	Medium
49	hospital	Rk Memorial Hospital	11.5975972, 78.5993824	30 km/h	Medium
50	hospital	Panneerselvam Hospital	11.598067, 78.599939	30 km/h	Medium
51	hospital	Sri Dhanvantari Children Hospital	11.5991194, 78.5999856	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
52	hospital	Government Hospital Attur	11.59709, 78.601343	30 km/h	Medium
53	hospital	Abhi Eye Hospital	11.5980207, 78.6003698	30 km/h	Medium
54	hospital	Government Hospital	11.597079, 78.600619	30 km/h	Medium
55	clinic	Dr. Ramdoss Clinic	11.595518, 78.600865	30 km/h	Medium
56	hospital	Arul Gandhi Physiotherapy Hospital	11.5979315, 78.6003262	30 km/h	Medium
57	hospital	St. Joseph Hospital	11.5990918, 78.6003644	30 km/h	Medium
58	clinic	G.V. Homeo Clinic	11.596099, 78.600999	30 km/h	Medium
59	hospital	Lakshitha Hospital	11.5952715, 78.6094785	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Vidya madir school	11.4356335, 78.1381408	30 km/h	Medium
9	college	Thiruvalluvar Govt arts college	11.4449352, 78.1595562	30 km/h	Medium
10	school	Venkateswara School	11.4468258, 78.1595761	30 km/h	Medium
11	school	Venkateswara Middle school	11.4472063, 78.1596191	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	school	Govt. Hr. Sec. School	11.4585324, 78.1933666	30 km/h	Medium
28	school	Rasi Matric HR Sec School	11.5760027, 78.5062808	30 km/h	Medium
30	school	Government Boys Higher Secondary School	11.598743, 78.5885472	30 km/h	Medium
31	school	Government Boys Higher Secondary School	11.599149, 78.5888931	30 km/h	Medium
60	school	MARUTHI PRIMAR, SECONDAR ,HIGHER SECONDARY SHOOL	11.5859471, 78.7186322	30 km/h	Medium
61	college	Maruthi Polytechnic College	11.5856852, 78.7201072	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43107, 77.88171



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39976, 77.88382



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39690, 77.88852



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38388, 77.89544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38386, 77.89881



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37820, 77.90198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41447, 78.08063



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41475, 78.08087



Coordinates: 11.43257, 78.12096



Coordinates: 11.45597, 78.18073



Coordinates: 11.45728, 78.18115



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46015, 78.18459



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.46034, 78.18528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45909, 78.18553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44994, 78.22115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44925, 78.22288



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44813, 78.22305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44767, 78.22372



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47202, 78.30442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49812, 78.32398



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50497, 78.33079



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50769, 78.33291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50860, 78.33951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51254, 78.34450



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51907, 78.34826



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54803, 78.35598



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58161, 78.41769



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58329, 78.44358



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57844, 78.48399



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57468, 78.49452



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57147, 78.49830



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57097, 78.50062



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57848, 78.51320



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57845, 78.51539



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58692, 78.54587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59228, 78.55728



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59744, 78.57347



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59550, 78.57463



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59533, 78.57596



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59889, 78.57708



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59974, 78.58060



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59965, 78.59643



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59891, 78.59656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59635, 78.60310



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59377, 78.60594



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59371, 78.61015



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.59503, 78.61074



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59314, 78.62100
